A COLLABORATION BETWEEN
ANDERSON SERANGOON JUNIOR COLLEGE &
NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 3

CHEMISTRY

9813/01

Paper 1

INSERT

21 September 2021

2 hours 30 minutes

INSTRUCTIONS

This insert contains information for Question 1. **Do not write your answers on the insert.**

This document consists of **3** printed pages and **1** blank page.

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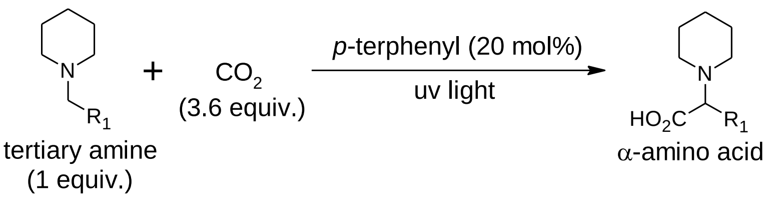
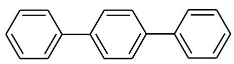
Information for Question 1

The advances in chiral catalysts to forge new carbon–carbon bonds with high levels of stereoselectivity have provided access to chiral molecular scaffolds that are ubiquitous in the fine-chemical and pharmaceutical industries.

Abstract 1 (H. Seo, M. H. Katcher, T. F. Jamison, Nat. Chem., 2016, 9, 453-456)

In the past decades, methods for carbon–carbon bond formation generally rely on two-electron mechanisms for carbon dioxide activation and require highly activated reaction partners. Alternatively, radical pathways could provide new reactivity under milder conditions. Here we demonstrate the direct coupling of carbon dioxide and amines to synthesise α-amino acids.





p-terphenyl ($C_{18}H_{14}$)

The proposed mechanism shown in Fig. 1.1 started with irradiation of p-terphenyl with ultraviolet light that produces the excited state of *p*-terphenyl*, which undergoes single-electron transfer (SET) with a tertiary amine to provide p-terphenyl radical anion and corresponding amine radical cation. The *p*-terphenyl$^{-\bullet}$ then reacts with CO_2 to form the CO_2 radical anion ($CO_2^{-\bullet}$). Concurrently, deprotonation of the amine radical cation affords the neutral α-amino radical. The key bond-forming step occurs by radical–radical coupling of neutral α-amino radical and CO_2 radical anion, followed by protonation to produce the desired α-amino acid.

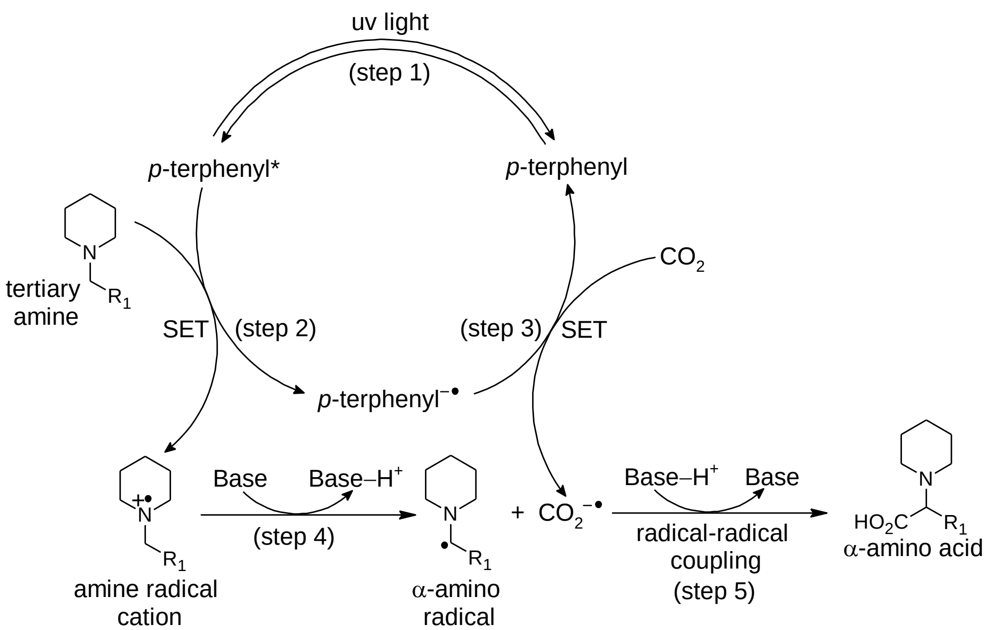


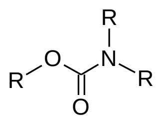
Fig. 1.1

The E (*p*-terphenyl / *p*-terphenyl$^{-\bullet}$) and E (CO_2 / $CO_2^{-\bullet}$) were measured with reference to saturated calomel electrode (SCE).

$$p\text{-terphenyl} + e^- \rightleftharpoons p\text{-terphenyl}^{-\bullet} \qquad E = -2.63 \text{ V with respect to SCE}$$
$$CO_2 + e^- \rightleftharpoons CO_2^{-\bullet} \qquad E = -2.21 \text{ V with respect to SCE}$$

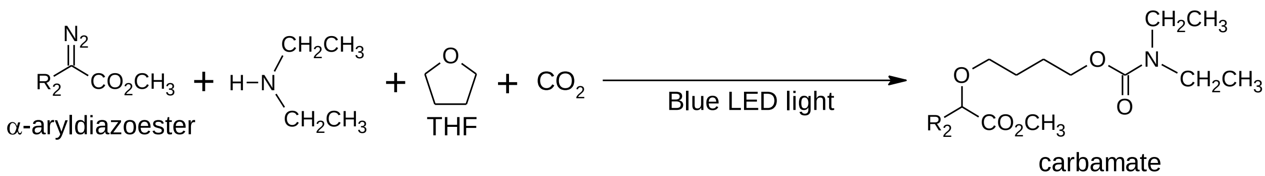
Abstract 2 (Cheng, R., Qi, C., Wang, L., Xiong, W., Liu, H., Jiang, H. **Green Chem.**, 2020, **22**, 4890-4895)

Organic carbamates represent an important class of compounds which often found in numerous natural products, pharmaceuticals and agrochemicals. However, traditional methods for producing organic carbamates rely on the use of toxic phosgene ($COCl_2$) or its derivatives as the raw material, thus are not environmentally friendly. A promising solution is to use carbon dioxide as a safe substitute for phosgene.



carbamate

An efficient and environmentally friendly method for the synthesis of organic carbamates from carbon dioxide has been developed using α-aryldiazoesters under mild conditions. A four-component coupling reaction involving aryl diazoesters, amines and carbon dioxide could occur rapidly and selectively, giving a variety of structurally diverse organic carbamates in moderate to high yields.



THF = Tetrahydrofuran LED = Light-Emitting Diode

A plausible mechanism shown in Fig. 1.2 is where a carbene intermediate is generated from α-aryldiazoester under the irradiation of blue LED light via an excited state α-aryldiazoester*. The carbene intermediate will be trapped by THF molecule to give an active oxonium ylide intermediate. The oxonium ylide intermediate then quickly react with a carbamate anion, giving an intermediate anion which then undergo protonation to yield the carbamate.

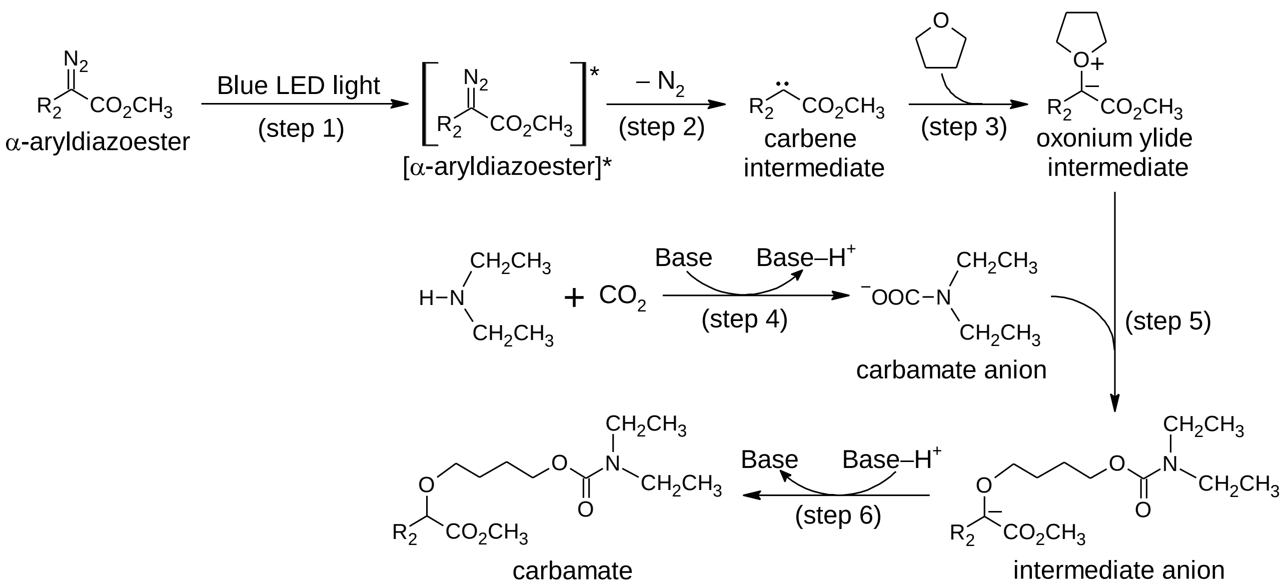


Fig. 1.2

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